AIAA Aviation Forum and ASCEND co-located Conference Proceedings
29 July - 2 August 2024, Las Vegas, Nevada
AIAA AVIATION FORUM AND ASCEND 2024

10.2514/6.2024-4117

Two and Three Dimensional Design and Characterization of Interchangeable Nozzles for a Supersonic Wind Tunnel Facility

Alexandre Mota*, Sarah Baird.†, Surabhi Singh‡ and William Engblom§
Embry-Riddle Aeronautical University, Daytona Beach, Florida, 32114

John Korte¶
Analytical Mechanics Associates, Hampton, Virginia, 23666

The aim of this study is to design and numerically verify interchangeable nozzles of equal length for application in a supersonic blowdown wind tunnel facility. The designed nozzles are intended to generate Mach 2, 3, and 4 flow across a 0.6096-meter (24-inch) test section with a square cross-sectional area of 0.1016 meters by 0.1016 meters (4 inches by 4 inches). The design of both the convergent (subsonic) and divergent (supersonic) sections of the nozzles were obtained through analytical codes, with special attention on minimizing the overall length of both sections without compromising flow uniformity by maintaining a zero wall slope at the inlet and exit planes. The design parameters were established, and viscous corrections to the designs were addressed. A comprehensive three-dimensional Computational Fluid Dynamics analysis was performed on all nozzles to ensure flow uniformity and to quantify the three-dimensional effects, such as corner vortices and "dog-bone" sidewall interactions, on the variation of test section parameters.

I. Introduction

Supersonic nozzle design is one of the most important design consideration for developing a supersonic wind tunnel facility. It requires maintaining a zero exit angle, or wall slope, at the exit where the nozzle joins the test section to eliminate the generation of a weak pressure wave. Maintaining such conditions can be challenging and requires better contour design criteria and numerical simulations to back these designs before fabricating them for installation in the wind tunnel. A typical algorithm for such a design is the method of characteristics (MOC), which utilizes the compatibility equations for the supersonic portion of the nozzle [1, 2]. MOC has been successful in designing both rocket nozzles [3] optimized for maximum thrust, and wind tunnel nozzles [4] optimized for uniform flow properties. Hence, it is important to define design requirements and desired flow characteristics when utilizing MOC.

It is possible to obtain successful nozzle designs without considering viscous effects. In previous studies, inviscid nozzle design has shown concurrence with Computational Fluid Dynamics (CFD) analysis [5]. However, boundary layer growth does have an impact on the attainable region of uniformity within the test section, as excessive boundary layer growth may lead to detachment from tunnel surfaces, disrupting the centerline flow. Unless at low stagnation pressures, the boundary layer is typically assumed to be turbulent in supersonic wind tunnels. Thus, it is beneficial to correct nozzle designs for boundary layer growth to obtain a wider region of uniform core flow in the test section.

The primary objective for the current work is to design and numerically validate interchangeable nozzles of same length for use in a supersonic blowdown-type wind tunnel facility. The nozzles designed in the current study are intended to produce Mach 2 and 3 flow, with provisions for potential Mach 4 use. The design parameters are laid out and viscous correction to the designs will be discussed. This work will explore design of both the convergent (subsonic) and divergent (supersonic) portion of the nozzles. A detailed three-dimensional CFD analysis will then be carried out on all the nozzles to ensure flow regularity and quantify the three-dimensional effects such as corner vortices and sidewall interaction on the variation of test section parameters, with the ultimate goal of limiting these variations as much as possible.

*Ph.D. Student, Aerospace Engineering. AIAA Student Member.
†Masters Student, Aerospace Engineering. AIAA Student Member.
‡Assistant Professor, Aerospace Engineering. AIAA Member.
§Professor, Aerospace Engineering. AIAA Associate Fellow.
¶SME Senior Principal, Aerospace Engineer. AIAA Associate Fellow.

II. Sivells' Method

As mentioned, the method of characteristics (MOC) is the algorithm typically used for designing the divergent portion of supersonic nozzles. Characteristic lines are defined as lines along which the derivatives of flow properties are indeterminate, and across which may be discontinuous. Along the characteristic lines, partial differential conservation equations can be combined in a way that ordinary differential equations are obtained, which are called compatibility equations. MOC consists of solving the compatibility equations step-by-step along the characteristic lines starting from the given initial conditions at some point or region in the flow. In this manner, the complete flow field can be mapped out along the general characteristics, and in the context of nozzle design, a contour can be obtained[1].

The code developed by J.C. Sivells utilizes MOC algorithms to generate ideal nozzle contours for supersonic and hypersonic wind tunnels based on provided input parameters [6]. In Sivells' approach, the axial velocity distribution is divided into three parts as illustrated in Fig. 1. In the initial region, from sonic point I to point E, the velocity distribution is characterized by a fourth-degree polynomial. In the intermediate section, from point E to point B, radial flow is assumed, taking place within the boundaries of characteristics EG and AB. Here, the contour from point G to point A forms a straight line at an angle ω relative to the axis. From flow separation considerations, ω is usually no more than 15° and a common value for many nozzles is 12°. Lastly, the Mach number distribution from point B to point C is described by a fifth-degree polynomial, and beyond point C, uniform flow is assumed. The coefficients of these polynomials are selected to ensure the continuity of the second derivative of axial velocity throughout the nozzle, with the condition of being zero at point C. Two characteristic solutions are needed: one for determining the throat contour from point H to point G, and the other for establishing the downstream contour from point A to point D, representing the theoretical end of the nozzle, with the characteristic CD being a straight line defining the onset of the uniform flow region. [7]

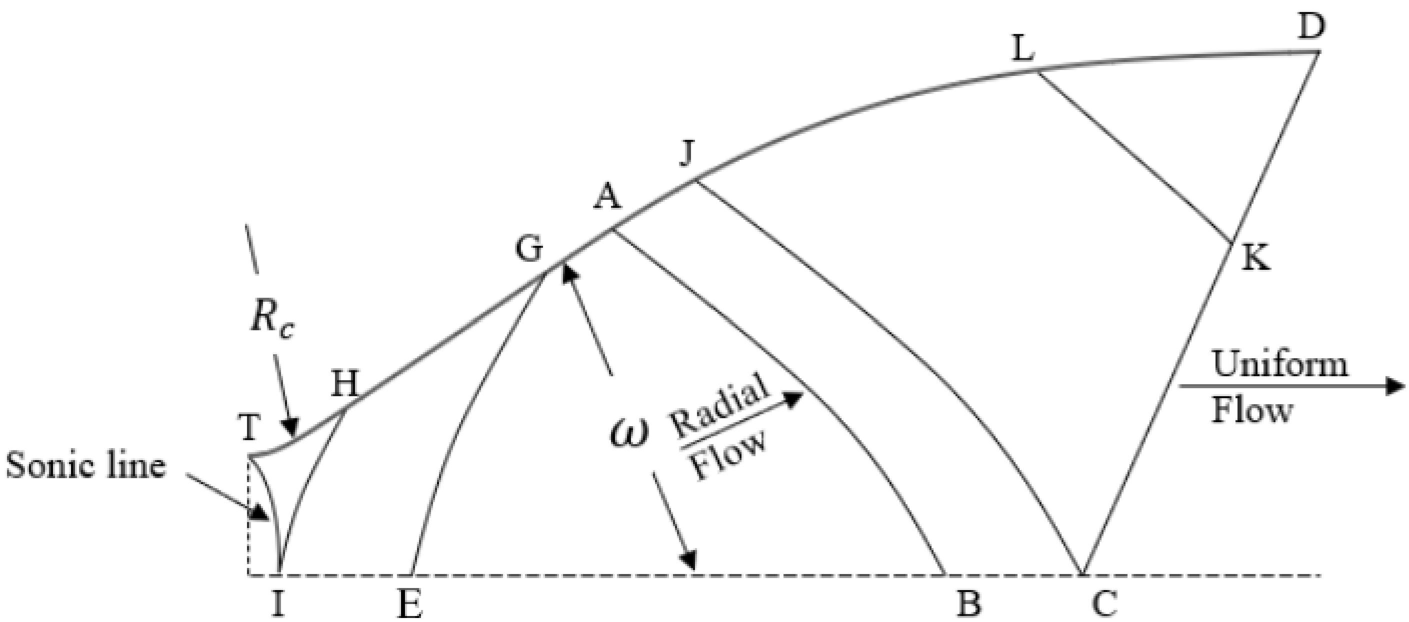


Fig. 1 Regions of inviscid contour.

Sivells' method relies on an initial estimation of the nozzle expansion angle and throat radius of curvature for a specified design Mach number, denoted as Mach at point C. The Mach number at point B is assumed to be 90 percent of the design Mach number. Using these initial inputs, Sivells' FORTRAN code can be used iteratively to calculate an estimated value for throat height, and consequently, nozzle area ratio, at specified stagnation conditions. The result is a viscous nozzle contour that has been corrected for some boundary layer development, however, the code's applied correction for boundary layer growth is greater than the displacement thickness on the contoured walls, which causes the flow to diverge in the test section[6]. Since the quality of flow in the test section is the main interest of wind tunnel nozzle design, Sivells' method requires additional corrections to properly accommodate boundary layer growth. First, a correction in Mach number is required to obtain the nozzle's 2-D inviscid contour for the previously obtained area ratio.

Then, to account for turbulent boundary layer development in the flow, 1-D theory is applied to estimate the flow's displacement thickness, δ^*. See Eq. 1.

$$\delta^* = y_{e_{des}} - y_s \cdot \left(\frac{A}{A^*}\right)_{1-D} \tag{1}$$

Where $y_{e_{des}}$ represents the desired exit height, y_s represents the throat height, and $\left(\frac{A}{A^*}\right)_{1-D}$ is the area ratio for design Mach number with $\gamma = 1.4$ - not the corrected area ratio obtained from Sivells' code. For a more accurate design, the displacement thickness in the middle of the test section can be verified through CFD simulations. The correction for boundary layer growth is made by calculating the corrected throat height and its corresponding area ratio for a desired exit height. This can be done by calculating the inviscid height through Eq. 2.

$$y_{inv} = y_{e_{des}} - 2 \cdot \delta^* \tag{2}$$

And subsequently the corrected throat height using Eq. 3.

$$y_{s_{cor}} = \frac{y_{inv}}{\left(\frac{A}{A^*}\right)_{1-D}} \tag{3}$$

The contour needs to be corrected for an area ratio such that Eq. 4 is satisfied.

$$\left(\frac{A}{A^*}\right)_{3-D} = \frac{y_{e_{des}}}{y_{s_{cor}}} \tag{4}$$

This correction will be translated as an increase in Mach number and nozzle length. Once completed, the final 3-D physical contour is obtained for the divergent section.

III. Hopkins-Hill Method

The convergent section of the nozzle is designed using the Hopkins-Hill method [8]. In their research, D. F. Hopkins and D. E. Hill developed a method to predict transonic flow features in nozzles with nearly sharp wall curvatures. The approach relies on a reference boundary designed to closely emulate the actual physical boundary of the nozzle. This reference boundary is mathematically expressed through an analytical function that ensures continuity in all derivatives. Serving as an initial approximation, the reference boundary is subject to iterative adjustments, refining its shape to better match the actual nozzle boundary in the throat region. The boundary's shape is modeled as a probability curve, determined by two parameters: C_1 and R_s. See Eq. 5.[8]

$$Y = C_1 \left[1 - \exp\left(\frac{-x^2}{2R_sC_1}\right)\right] + y_s \tag{5}$$

The parameter R_s sets the throat's radius of curvature, while C_1 determines the entrance angle.Varying these parameters creates a family of curves, some depicted in Fig. 2.

Normalizing distances by throat height, y_s, introduces the parameter H_r, representing the reference boundary's contour in terms of C_1 and R_s (see Eq. 6).

$$H_r = C_1 \left[1 - \exp\left(\frac{-\xi^2}{2R_sC_1}\right)\right] + 1 \tag{6}$$

Where ξ is the normalized distance in the x-axis. The first x-point of the distribution curve on the centerline axis, ξ_1 is obtained from R_s and the inlet's entrance angle θ_{sub}. See Eq. 7 below.

$$\xi_1 = -R_s \tan(\theta_{sub}) \exp(0.5) \tag{7}$$

The value of C_1 can be calculated using Eq. 8.

$$C_1 = \frac{\xi_1^2}{R_s} \tag{8}$$

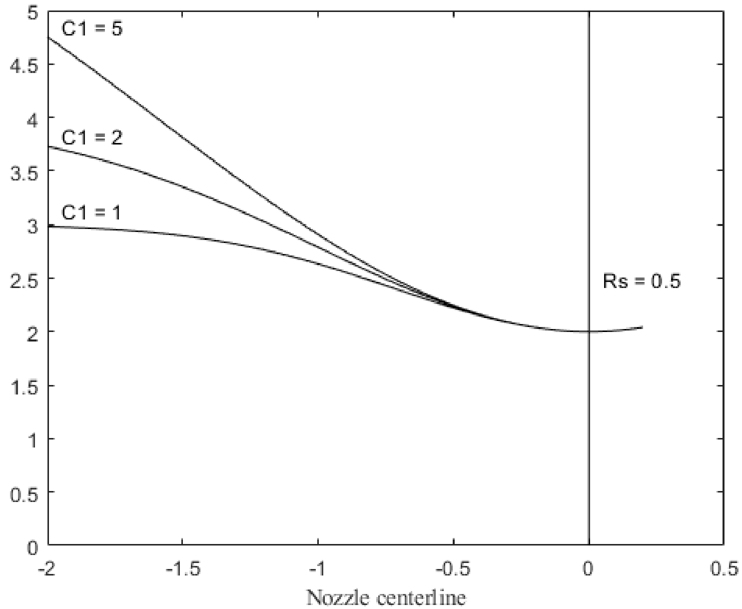


Fig. 2 Reference boundary shape.

The subsequent ξ-points along the distribution curve are computed based on the distance between ξ and the throat, where $\xi = 0$, and the desired number of points, denoted as n_{pts}. This parameter influences the slope of the contour. The x-axis distribution is expressed by Eq. 9 below.

$$\xi_{n+1} = \xi_1 + n \cdot dx \tag{9}$$

Here, dx is determined by Eq. 10, given by:

$$dx = \frac{\xi_1}{n_{pts}} \tag{10}$$

In this fashion, as n approaches n_{pts}, the reference boundary aligns with the throat at $\xi = 0$, resulting in $H_r = 1$ with $y = y_s$. For an entrance angle of 45 degrees, this method establishes a ramp contour for the convergent section of the nozzle, which is critical for length considerations. From transitional R_{e_x}, a minimum length can be calculated so that turbulent flow can be effectively induced into the throat. See Eq. 11.

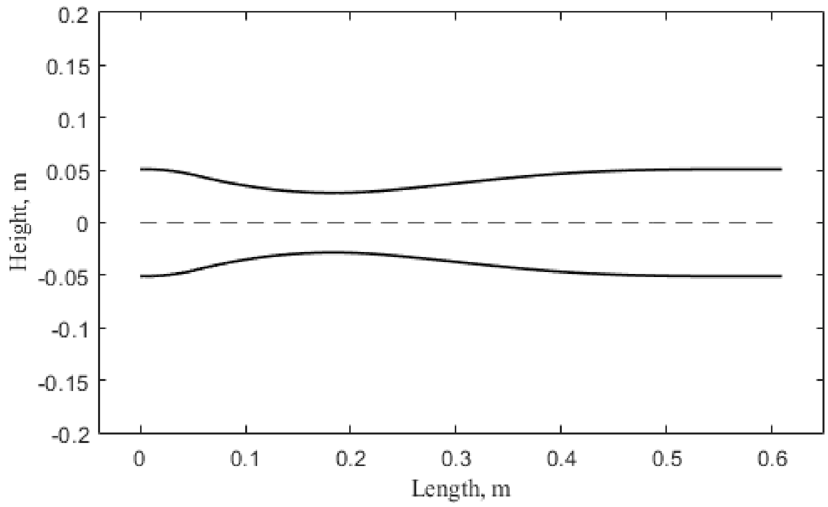
$$L_{min} = \frac{R_{e_x} \nu}{U} \tag{11}$$

This design adjustment is crucial for achieving the intended flow characteristics of the nozzle. The inlet can be curve fitted to a circular form at a desired height provided that the contour's slope remains relatively constant.

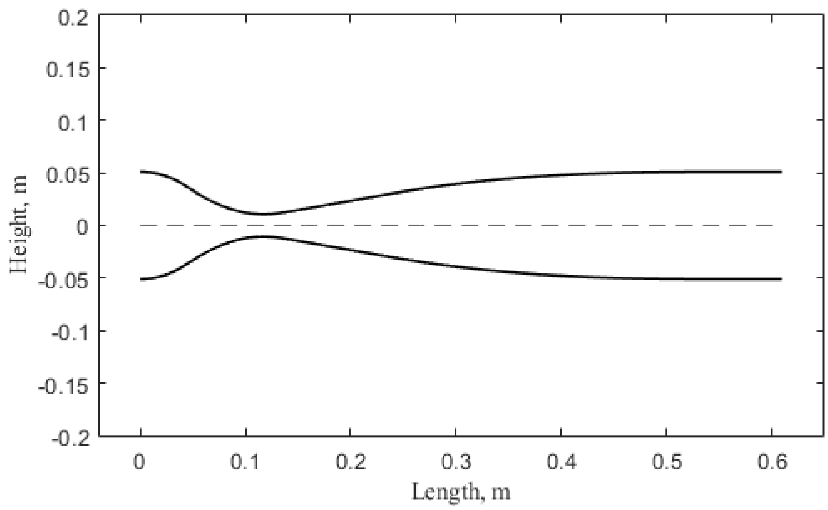
IV. Nozzle Design Criteria

The nozzles to be designed are nozzles of different operating design Mach numbers, to be used interchangeably in a supersonic blowdown type facility without altering the relative location of the wind tunnel test section. This means that they will all have to be of the same nominal length. The three operating Mach numbers desired are Mach 2, 3 and 4. Hence, Mach 4 nozzle contour was used as the main length determining criteria.

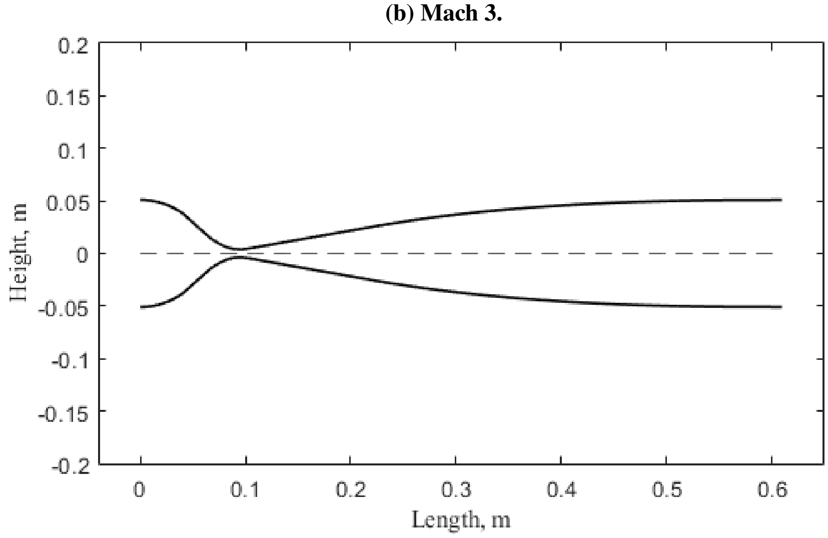
The blowdown tunnel is planned to have a square test section of cross-section of 0.1016 meters by 0.1016 meters (4 inches by 4 inches) which sets the half-height of the nozzle at 0.0508 meters (2 inches). Table 1 mentions additional operating parameters set for the nozzle design. The final contours obtained as a result of this design process are shown in Figs. 3a, 3b and 3c for the Mach 2, 3, and 4 nozzles, respectively.



(a) Mach 2.



(b) Mach 3.



(c) Mach 4.

Fig. 3 Nozzle shape.

Table 1 Nozzle Design Parameters

Nozzle 1	Uniform Mach 2 flowfield at nozzle exit Dry air supply at P_0 of 2.4 - 3.8 $\times 10^5$ Pa (35 - 55 psiA) and T_0 of 294.4 K (530 R)
Nozzle 2	Uniform Mach 3 flowfield at nozzle exit Dry air supply at P_0 of 5.5 - 8.3 $\times 10^5$ Pa (80 - 120 psiA) and T_0 of 294.4 K (530 R)
Nozzle 3	Uniform Mach 4 flowfield at nozzle exit Dry air supply at P_0 of 10.3 - 12.4 $\times 10^5$ Pa (150 - 180 psiA) and T_0 of 294.4 K (530 R) The downstream pressure will be reduced by external means
Inlet Geometry	0.1016 m × 0.1016 m (4 in × 4 in)
Exit (Also Test Section) Geometry	0.1016 m × 0.1016 m (4 in × 4 in)
Wall Type	Parallel walls
Additional Features	All nozzles will have common inlet and exit All nozzles will have the same length

V. Grid Topology and Sensitivity Study

Grid information can be seen below in Table 2 for Mach 2, 3 and 4 nozzles. Due to a larger area ratio, and consequently a more contoured surface, the mesh grid for Mach 4 nozzle required higher cell count than Mach 2 and 3 surfaces. Since the initial grid for Mach 4 nozzle was already much finer than the other nozzles, the percentage increase in cell count for the finer grids was not as high as it was for Mach 2 and 3. Overall, Mach 4 mesh grid ranged from two to three times more cells than Mach 2 and 3 meshes.

Table 2 Grid cell count and percentage increase

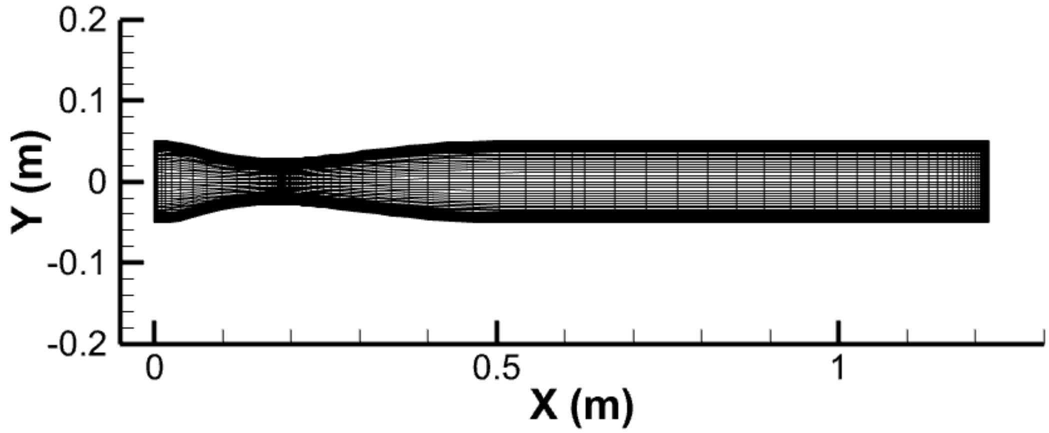
	Coarse	Middle	Fine
Mach 2	536,648	960,498 (78.98 %)	1,759,688 (83.21 %)
Mach 3	536,648	960,498 (78.98 %)	1,759,688 (83.21 %)
Mach 4	1,891,248	2,480,172 (31.14 %)	3,228,012 (30.15 %)

The software used for grid generation was Fidelity Pointwise. The finer mesh grid topology can be seen below in Figs. 4a, 4b and 4c for all nozzles. Figs. 5a, 5b and 5c display throat clustering grid shots. The meshes distribution in the YZ plane can be seen in Figs. 6a, 6b and 6c.

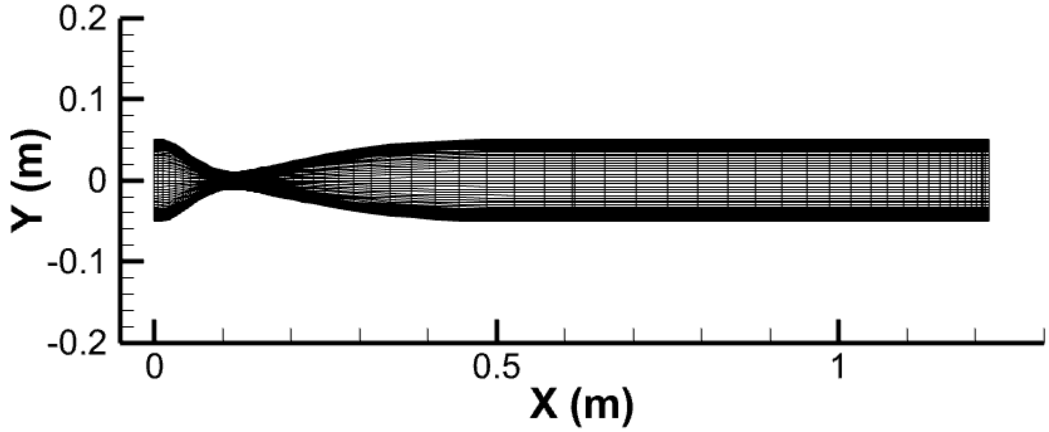
CFD parameters are given in Table 3. The flow channel includes the 3-D nozzle, as well as a 0.6096 m (24 in) test section added to examine the streamwise variation in Mach number as the boundary layer thickens through this region.

Table 3 CFD Solver Parameters

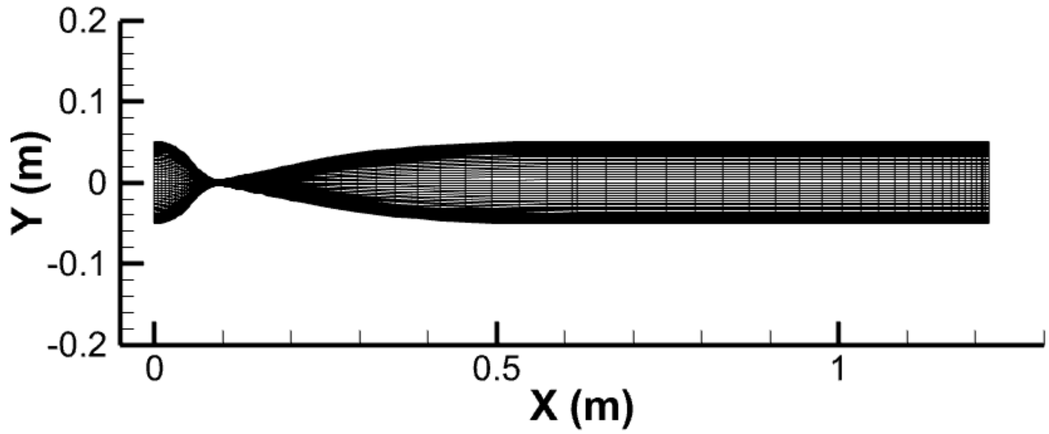
Boundary conditions	Absolute total pressure inlet: Mach 2: 310,275 Pa (45 psiA) Mach 3: 689,500 Pa (100 psiA) Mach 4: 1,137,675 Pa (165 psiA)
	Static pressure outlet: $p_b = p_e$ (perfectly expanded)
	Wall: Nozzle internal surfaces
Submodels	Viscous model: k-omega SST
Numerical Scheme	AUSM
Spatial Discretization	Turbulent kinetic energy, specific dissipation rate: Second Order Upwind



(a) Mach 2.

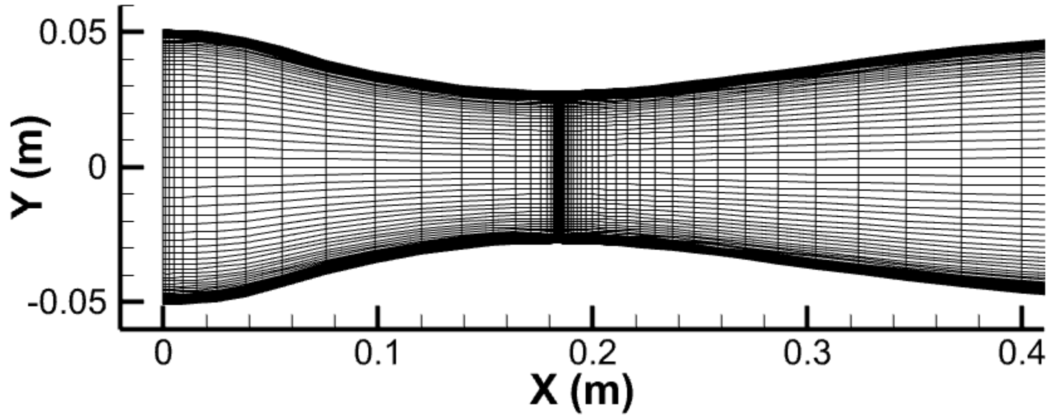


(b) Mach 3.

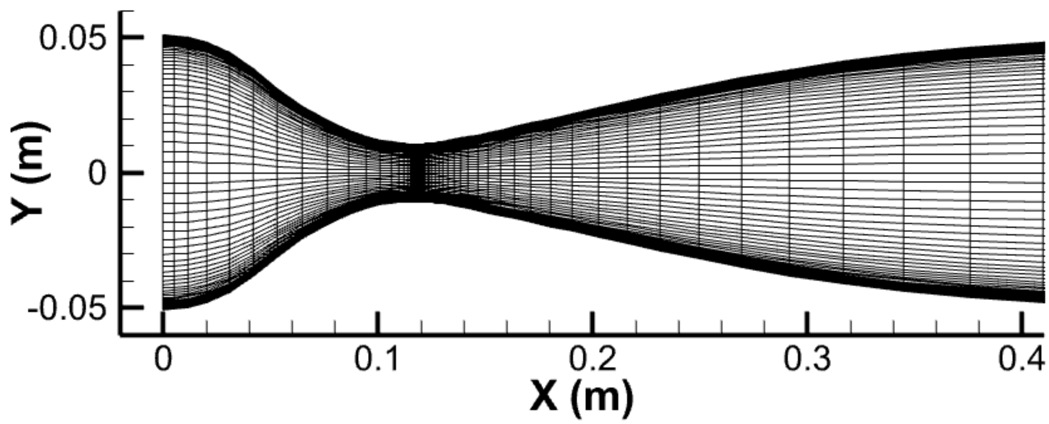


(c) Mach 4.

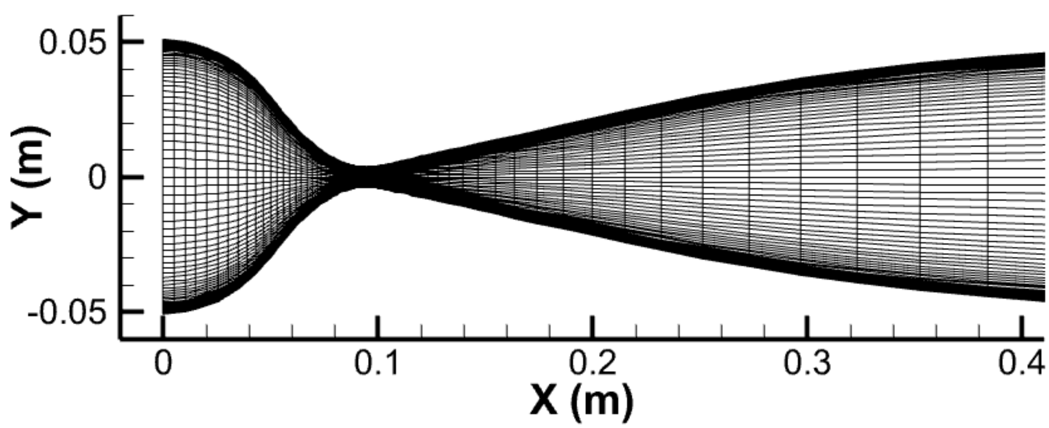
Fig. 4 Full channel mesh grid.



(a) Mach 2.

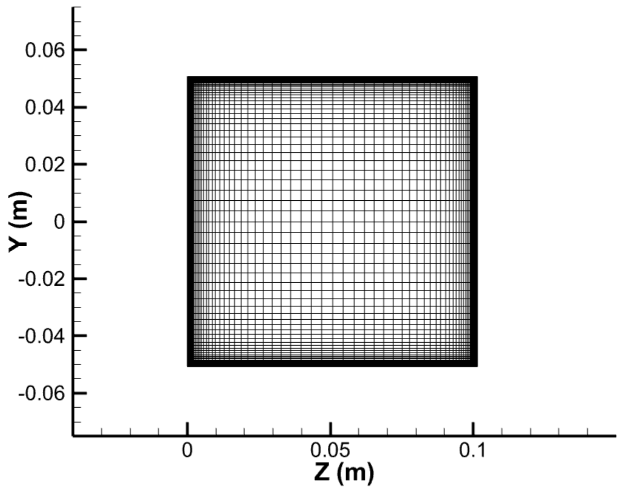


(b) Mach 3.

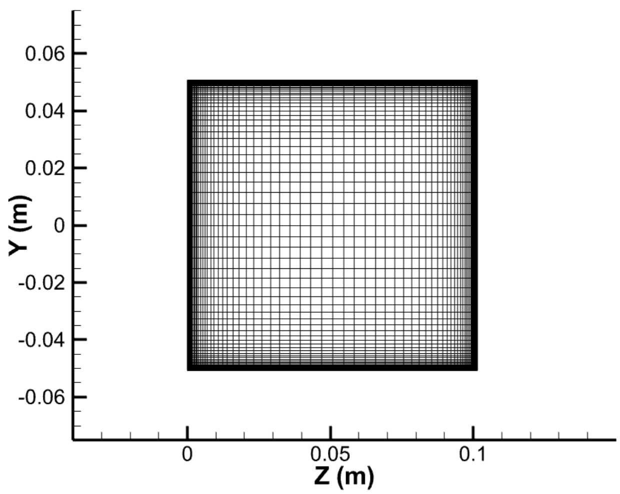


(c) Mach 4.

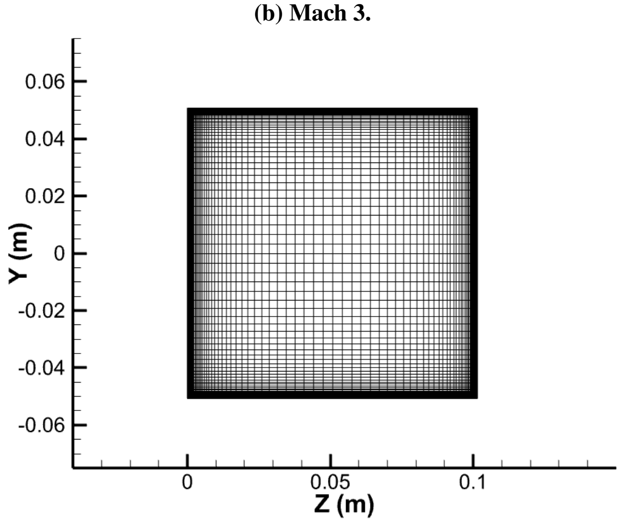
Fig. 5 Throat clustering mesh grid.



(a) Mach 2.



(b) Mach 3.

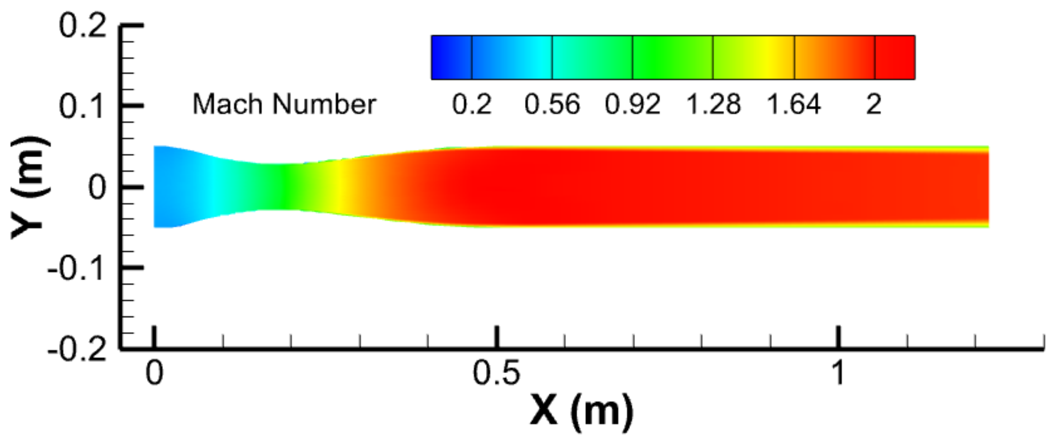


(c) Mach 4.

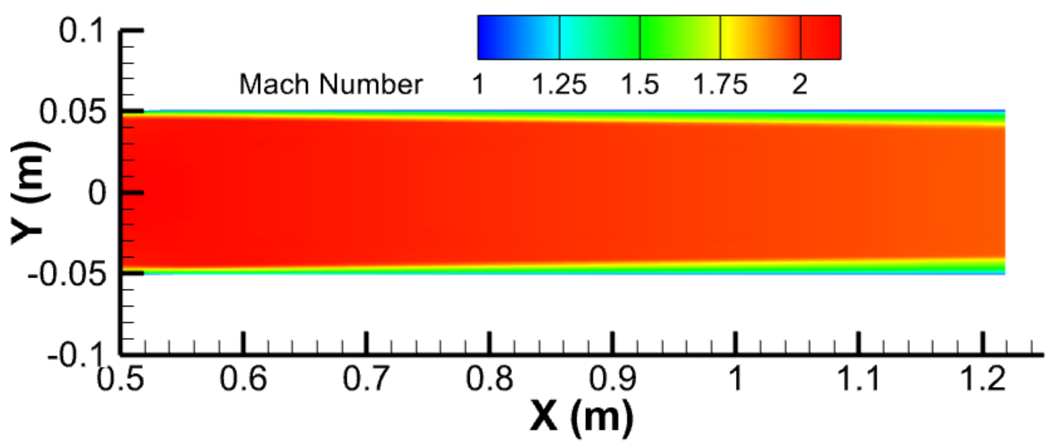
Fig. 6 Mesh grid in the YZ plane.

VI. Results

Computational results were obtained in Ansys Fluent for all nine meshes using an implicit density-based AUSM formulation with a 2nd order k-omega SST turbulence model. The grid sensitivity study was successful, and numerical results display good agreement with expected values. The finer mesh grid results exhibited an expansion up to Mach 2.019, 3.054 and 4.117, visible in the contours of Mach number taken at the z-axis centerline shown in Figs. 7, 8 and 9 respectively. The Mach 2 nozzle exhibits greater fluctuations in Mach number throughout the test section compared to isentropic expectations with a total fluctuation of 3.57%, ranging from -2.60% at the nozzle exit plane to -6.17% at the test section exit plane. Mach 3 had a total fluctuation of 2.59%, ranging from -2.24% to -4.83%, and Mach 4 had a total fluctuation of 2.99%, from -3.13% to -6.12% at the nozzle and test section exit planes, respectively. This larger variation in Mach number may be partially attributed to the smaller three-dimensional correction requirement for the Mach 2 nozzle, based on the one-dimensional isentropic area ratio. As the Mach number increases, the one-dimensional isentropic area ratio becomes more extrapolated and less accurate when compared to actual three-dimensional flow. This inaccuracy results in a larger displacement thickness (δ^*) correction, as described by Eq. 1.



(a) Nozzle + test section.

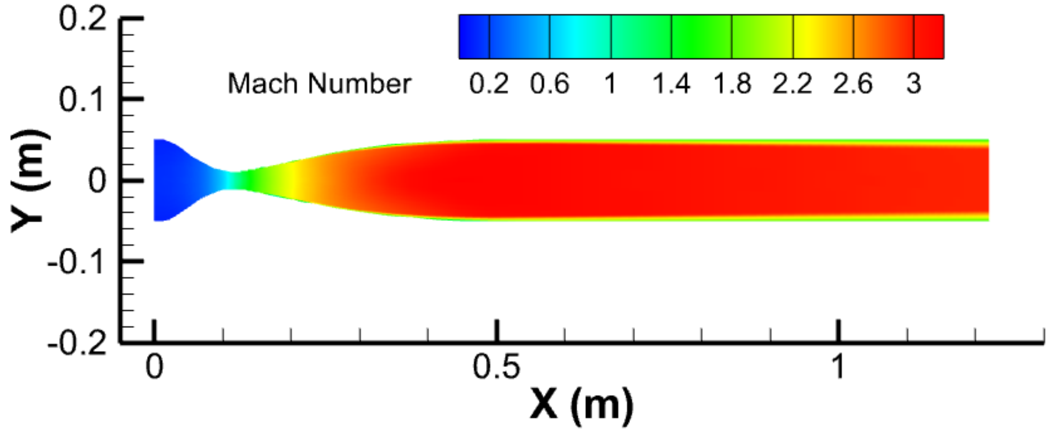


(b) Test section. ($x = 0.61\,m$)

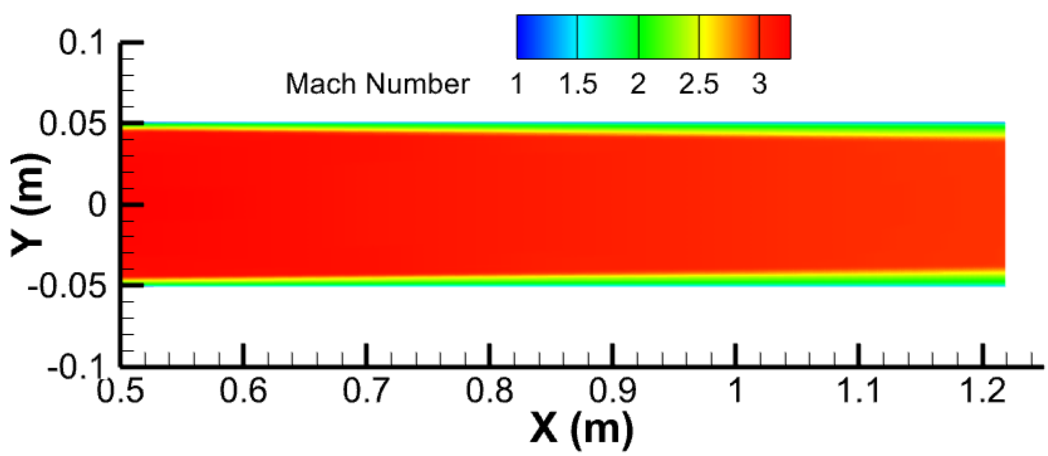
Fig. 7 Streamwise Mach contour on the x-y plane for Mach 2 nozzle ($p_0 = 45\,psia$).

Figs. 10a, 10b, and 10c display Mach number variation with nozzle height at the nozzle exit plane. The curves are consistent across all three nozzles, indicating that the Mach number remains constant over most of the y-axis distance.

Figs. 11, 12 and 13 show cross-sectional cuts of Mach number contours at the nozzle and test section exit planes. Boundary layer growth at the nozzle planar walls is evidently larger than at the contoured surfaces, generating what is commonly called "dog-bone" contours, and which are shown to become more pronounced with higher Mach numbers. This excessive growth in boundary layer decelerates the flow near the walls, limiting the core flow region of desired Mach number. Therefore, as the Mach number increases, designing planar nozzles becomes more challenging, indicating the

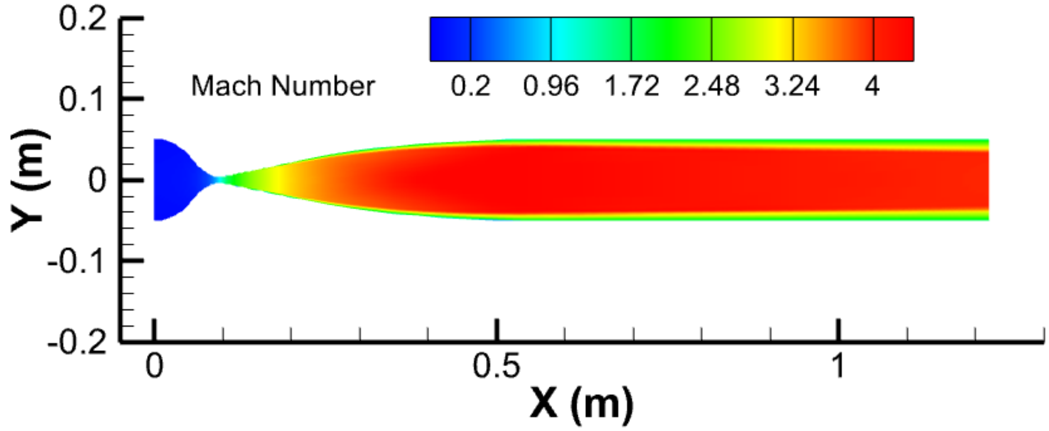


(a) Nozzle + test section.

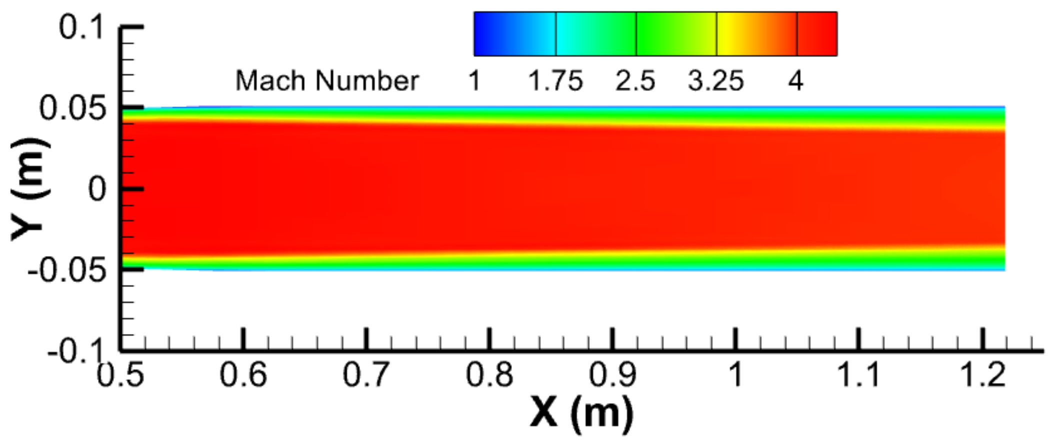


(b) Test section. ($x = 0.61\,m$)

Fig. 8 Streamwise Mach contour on the x-y plane for Mach 3 nozzle ($p_0 = 100\,psia$)**.**

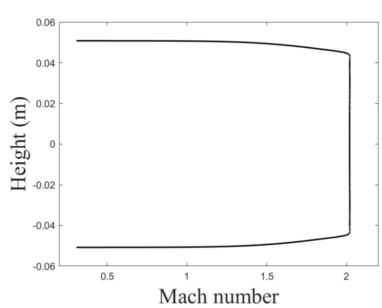


(a) Nozzle + test section.

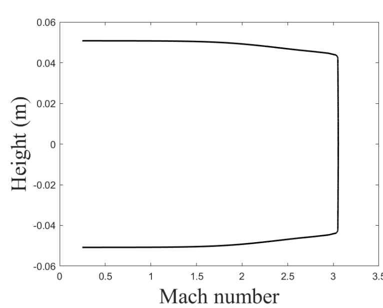


(b) Test section. ($x = 0.61\,m$)

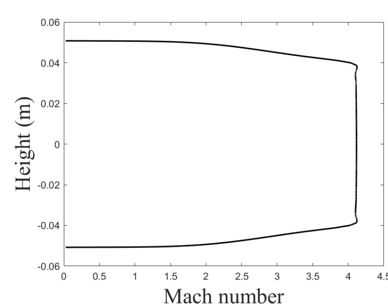
Fig. 9 Streamwise Mach contour on the x-y plane for Mach 4 nozzle ($p_0 = 165\,psia$)**.**



(a) Mach 2 nozzle



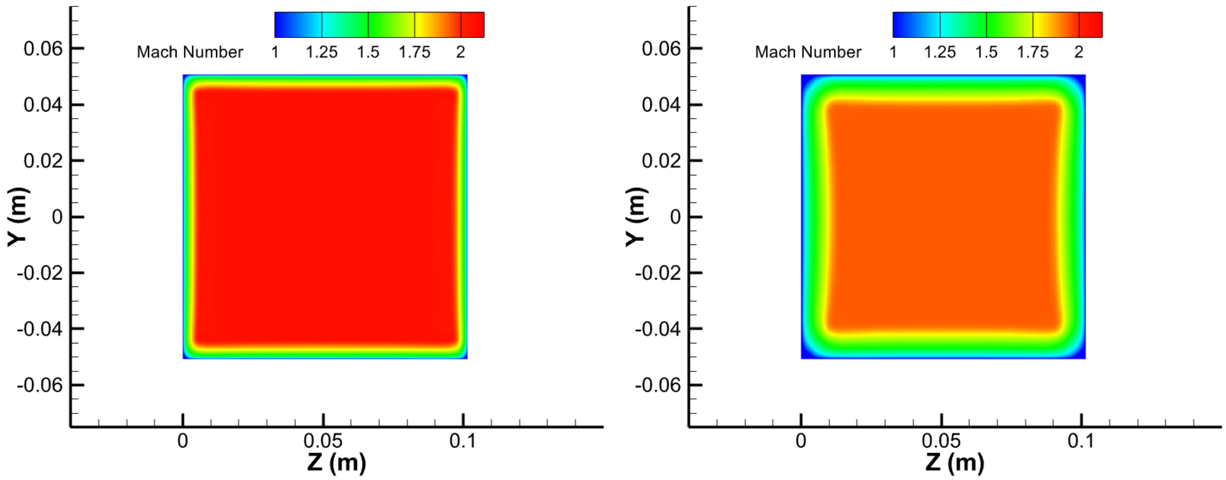
(b) Mach 3 nozzle



(c) Mach 4 nozzle

Fig. 10 Variation in Mach number at nozzle exit plane.

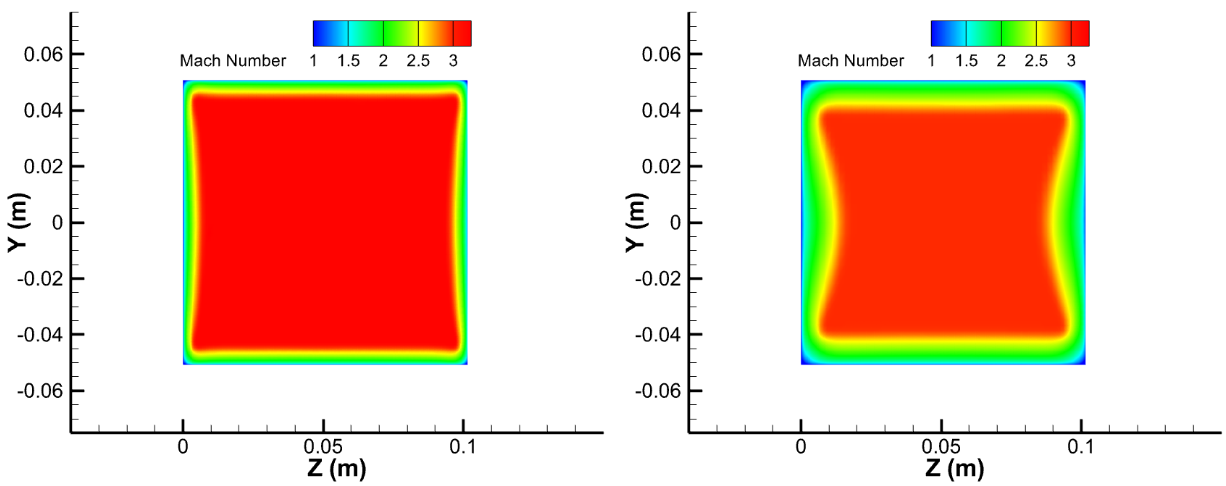
need to consider axisymmetric nozzles to alleviate this effect.



(a) **End of nozzle.** ($x = 0.61\,m$)

(b) **End of test section.** ($x = 1.21\,m$)

Fig. 11 Streamwise Mach contour on the y-z plane for Mach 2 nozzle ($p_0 = 45\,psia$).



(a) **End of nozzle.** ($x = 0.61\,m$)

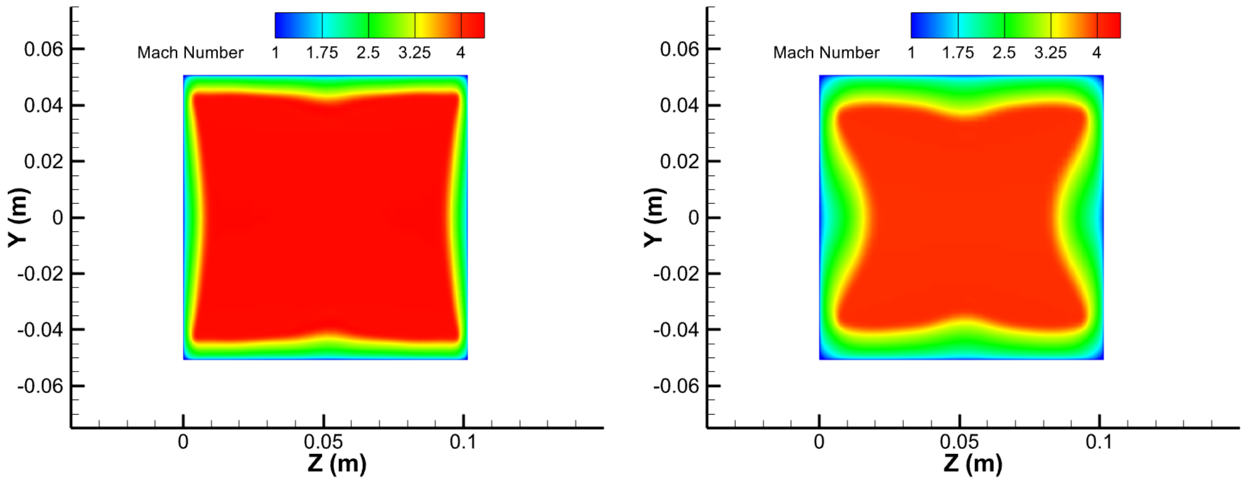
(b) **End of test section.** ($x = 1.21\,m$)

Fig. 12 Streamwise Mach contour on the y-z plane for Mach 3 nozzle ($p_0 = 100\,psia$).

Comparison of analytical isentropic results and computational results taken from the center point at the nozzle and test section exit planes are tabulated in Table 4. Due to small levels of unsteadiness in the solution, some stagnation values are shown to be slightly above/below the isentropic target value. Furthermore, Table 4 shows higher numerical exit plane static properties across the test section compared to those predicted under ideal conditions. This indicates that the system performance is not entirely isentropic for all nozzles. Overall, this three-dimensional result validates the chosen nozzle designs by exhibiting minimal test section Mach number variation and promoting flow uniformity by limiting the strength of shocks in this region.

VII. Conclusion

A combination of Sivells' and Hopkins-Hill's method was successfully employed to generate the supersonic and subsonic sections of the nozzle contours, respectively. Combining both techniques is crucial for achieving a consistent nozzle length suitable for high-speed flow, independent of the Mach number. The nozzle designs were validated



(a) End of nozzle. (x = 0.61 m)

(b) End of test section. (x = 1.21 m)

Fig. 13 Streamwise Mach contour on the y-z plane for Mach 4 nozzle (p_0 = 165 $psia$).

Table 4 Numerical and Isentropic Test Section Parameters

		Mach 2		Mach 3		Mach 4	
		Isentropic expectation	Test section range	Isentropic expectation	Test section range	Isentropic expectation	Test section range
M_{TS}		2.073	1.945 - 2.019	3.124	2.973 - 3.054	4.250	3.990 - 4.117
p_0	(psia)	45	44.84 - 44.87	100	101.1 - 101.2	165	165.5 - 165.6
p_{TS}	(psia)	5.133	5.566 - 6.242	2.263	2.535 - 2.867	0.7829	0.9322 - 1.103
T_{TS}	(K)	168.3	162.2 - 167.6	99.73	103.1 - 106.8	63.8	67.2 - 70.5
T_0	(K)	294.4	294.1 - 294.1	294.4	295.0 - 295.1	294.4	294.4 - 294.4
ρ_{TS}	(kg/m^3)	0.7327	0.8244 - 0.8948	0.5451	0.5905 - 0.6448	0.2948	0.3333 - 0.3758
u_{TS}	(m/s)	539.1	504.6 - 515.3	625.4	621.5 - 615.7	680.5	671.3 - 676.3
a_{TS}	(m/s)	260.0	255.2 - 259.4	200.2	203.5 - 207.1	160.1	164.3 - 168.3

using CFD simulations. It becomes evident that three-dimensional "dog-bone" effects, which decelerate flow near the walls, intensify with higher Mach numbers. This highlights the need for more substantial adjustments in the design of high-Mach nozzles. Additionally, computational results show that boundary layer growth is more pronounced on planar walls of the nozzles compared to contoured walls, consistent with real-flow phenomena. Overall, the Mach 2, 3, and 4 nozzles designed perform adequately for the planned blowdown wind tunnel facility. However, observed three-dimensional effects suggest that beyond a certain Mach number, it may be beneficial to consider designing axisymmetric nozzles to enhance flow quality in the test section.

References

[1] Anderson, J. D., *Modern compressible flow: with historical perspective*, Vol. 12, McGraw-Hill New York, 1990.

[2] McCabe, A., “Design of a supersonic nozzle,” 1964.

[3] Rao, G., and Beck, J., “Use of discontinuous exit flows to reduce rocket nozzle length,” *30th Joint propulsion conference and exhibit*, 1994, p. 3264.

[4] Plyashechnik, V., Byrkin, A., and Verkhovsky, V., “Gasdynamic design of shaped nozzles for supersonic wind tunnels, allowing for viscosity,” *West-East High Speed Flow Field Conference, Moscow, Russia*, Vol. 11, 2007, pp. 19–22.

[5] Korte, J., "Inviscid design of hypersonic wind tunnel nozzles for a real gas," *38th Aerospace Sciences Meeting and Exhibit*, 2000, p. 677.

[6] Sivells, J. C., "A computer program for the aerodynamic design of axisymmetric and planar nozzles for supersonic and hypersonic wind tunnels," *US Air Force Arnold Engineering Development Center Rept. AEDC-TC-78-63, Arnold AFB, TN*, 1978.

[7] Sivells, J. C., "Aerodynamic design of axisymmetric hypersonic wind-tunnel nozzles," *Journal of Spacecraft and Rockets*, Vol. 7, No. 11, 1970, pp. 1292–1299.

[8] Hopkins, D., "Effect of small radius of curvature on transonic flow in axisymmetric nozzles." *AIAA Journal*, Vol. 4, No. 8, 1966, pp. 1337–1343.